PANEL DEGRADATION DETECTION SYSTEM
Omni-Grid Renewable Energy Intelligence Platform

1. Model Overview

Attribute	Details
Model Name	SolarShield-Degradation v1.0
Model Type	Weather-Normalised Performance Degradation Detection
Primary Algorithm	XGBoost Regression + Residual Trend Analysis
Secondary Components	Rolling Trend Detection + Statistical Thresholding
Version	1.0.0
Last Updated	May 2026
Owner	AIROBOD Data Science Team
Status	Design Phase

2. Problem Statement

Solar photovoltaic systems gradually lose efficiency over time due to multiple environmental and operational factors. Traditional monitoring systems often fail to distinguish between temporary weather-related performance fluctuations and genuine panel degradation.

This model is designed to continuously monitor solar asset performance and detect long-term degradation patterns by comparing actual energy generation against weather-normalised expected generation.

The system aims to provide early detection of:

- Panel degradation
- Dust and soiling accumulation
- Shading effects
- Micro-cracks
- Hotspot formation
- Electrical mismatch losses

- Aging-related efficiency decline
- Wiring or connection losses

The model enables proactive maintenance planning and improves long-term asset performance management.

3. Intended Use

Primary Use Cases

1. Detect gradual solar panel degradation
2. Identify persistent underperformance
3. Monitor panel/string efficiency trends
4. Trigger cleaning or inspection recommendations
5. Improve operational efficiency of solar plants
6. Support asset lifecycle management

Target Users

- Solar plant operators
- Asset management teams
- Maintenance planners
- Reliability engineering teams
- Grid operations teams
- Renewable energy analytics teams

Out-of-Scope Uses

- Real-time inverter fault detection
- Battery degradation analysis
- Wind turbine monitoring
- Grid frequency instability analysis
- Short-term generation forecasting

4. Model Architecture

System Workflow

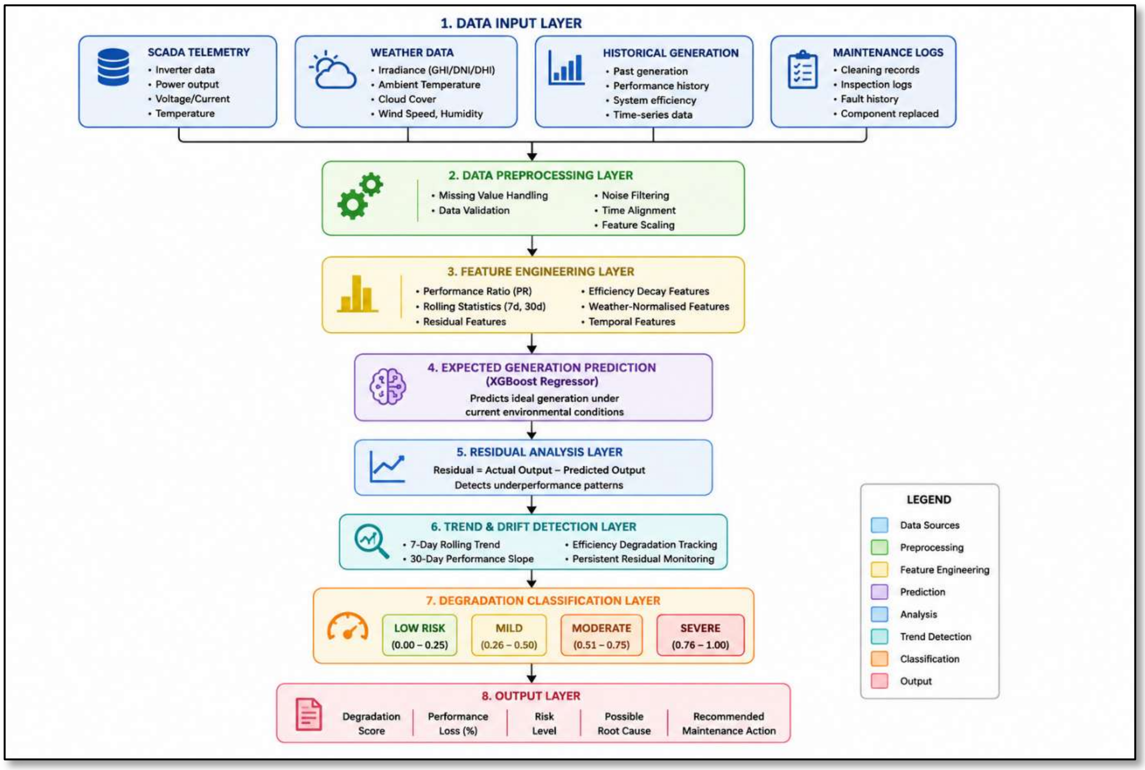

Figure 1 illustrates the end-to-end workflow architecture of the proposed SolarShield-Degradation Detection System.

The Panel Degradation Detection System follows a multi-stage machine learning workflow designed to identify long-term solar panel underperformance and degradation patterns.

The architecture begins with the Data Input Layer, where data is collected from multiple sources including SCADA telemetry systems, weather APIs, historical generation records, and maintenance databases. These sources provide both real-time and historical operational context for the solar assets.

The incoming data is then processed within the Data Preprocessing Layer. This stage performs missing value handling, noise reduction, timestamp synchronization, feature scaling, and validation checks to ensure data quality and consistency before model inference.

Next, the Feature Engineering Layer transforms the raw telemetry into higher-level analytical features. This includes performance ratio calculations, rolling statistical indicators, residual-based features, temporal encodings, and weather-normalised operational metrics. These engineered features help the model distinguish between environmental fluctuations and genuine degradation behaviour.

The processed features are then passed into the Expected Generation Prediction Layer, which uses an XGBoost Regressor to estimate how much power the solar system should generate under ideal operating conditions given the current weather and site parameters.

Following this, the Residual Analysis Layer compares the predicted generation against the actual generation output. The residual values help identify persistent energy losses and abnormal efficiency decline patterns.

The Trend and Drift Detection Layer then evaluates long-term operational behaviour using rolling averages, degradation slopes, and persistent residual monitoring. This stage is critical because solar panel degradation typically occurs gradually over extended periods rather than as sudden failures.

The resulting degradation indicators are then processed by the Degradation Classification Layer, where the system categorises the asset into different severity levels such as Low Risk, Mild, Moderate, or Severe Degradation.

Finally, the Output Layer generates actionable operational insights including degradation scores, estimated performance loss percentages, possible root causes, and recommended maintenance actions. These outputs can be integrated into operational dashboards, maintenance scheduling systems, and renewable energy asset management platforms.

5. Core Modelling Approach

5.1 XGBoost Regression Model

The primary regression model predicts the expected solar energy generation under normal operating conditions using environmental, temporal, and operational variables.

The model learns the relationship between:

- Irradiance
- Temperature
- Weather conditions
- Time-based generation patterns
- Historical site behaviour

and predicts the generation that the system should theoretically produce.

5.2 Residual Analysis

Residual values are calculated as:

Residual = Actual Generation - Predicted Generation

Persistent negative residuals indicate underperformance.

Residual analysis helps identify whether energy losses are caused by:

- panel degradation

- soiling
- shading
- efficiency loss
- hardware deterioration

rather than temporary weather fluctuations.

5.3 Rolling Trend Detection

Because degradation occurs gradually, rolling statistical analysis is used to detect long-term efficiency decline.

The system analyses:

- 7-day rolling performance ratio
- 30-day degradation slope
- long-term efficiency decay rate
- rolling generation loss percentage

This allows the model to distinguish between:

- short-term anomalies
- long-term degradation behaviour

6. Training Data

Attribute	Details
Data Sources	Existing Omni-Grid SCADA + Weather Pipeline
Time Period	2022 – 2026
Geographic Coverage	NSW, QLD, VIC, SA, WA
Solar Sites	Utility-scale and commercial solar plants
Resolution	5-minute and hourly aggregated data
Dataset Type	Real + synthetic degradation scenarios

Data Components

SCADA Telemetry

- AC power output

- DC voltage/current
- inverter efficiency
- panel temperature
- system efficiency
- operational status

Weather Data

- irradiance (GHI/DNI/DHI)
- ambient temperature
- cloud cover
- humidity
- wind speed
- solar elevation angle

Historical Performance Data

- historical generation
- expected performance ratio
- rolling averages
- seasonal generation behaviour

7. Synthetic Degradation Simulation

Since real labelled degradation events are limited, synthetic degradation patterns are generated to simulate realistic panel deterioration.

Simulated Scenarios

Gradual Aging

Efficiency decline over months

98% → 96% → 93% → 89%

Dust/Soiling Accumulation

- progressive reduction in generation
- increased panel temperature
- reduced performance ratio

Shading Effects

- partial daytime output suppression
- intermittent generation drops

Electrical Degradation

- voltage mismatch
- current imbalance
- persistent underperformance

8. Input Features

Feature Category	Input Features
Weather Features	Global Horizontal Irradiance (GHI), Direct Normal Irradiance (DNI), Diffuse Horizontal Irradiance (DHI), Ambient Temperature, Cloud Cover, Humidity, Wind Speed, Solar Elevation Angle
Operational Features	Actual Power Output, Expected Power Output, Inverter Efficiency, Panel Temperature, System Efficiency, DC Voltage, DC Current, AC Power Output
Temporal Features	Hour of Day, Day of Week, Month, Season, Day Length
Engineered Features	Performance Ratio (PR), Rolling 7-Day PR Mean, Rolling 30-Day PR Trend, Efficiency Decay Rate, Residual Trend, Temperature Stress Index, Days Since Cleaning, Days Since Maintenance, Long-Term Generation Drift

9. Output Specification

Example Output

{

"site_id": "SOLAR-NSW-023",

"panel_group_id": "STRING-12",

"timestamp": "2026-05-07T12:00:00Z",

"predicted_generation_kw": 284.2,

"actual_generation_kw": 256.7,
"performance_loss_percent": 9.7,
"degradation_score": 0.81,
"risk_level": "HIGH",
"possible_issue": "PANEL_SOILING_OR_AGING",
"recommended_action": "INSPECT_AND_CLEAN_WITHIN_7_DAYS"
}

10. Degradation Severity Levels

Score Range	Risk Level	Description
0.00 – 0.25	Low	Normal operation
0.26 – 0.50	Mild	Minor efficiency decline
0.51 – 0.75	Moderate	Persistent underperformance
0.76 – 1.00	High	Significant degradation detected

11. Evaluation Metrics

Regression Metrics

MAE (Mean Absolute Error)

Measures average forecasting error magnitude.

Used to evaluate expected generation prediction quality.

RMSE (Root Mean Squared Error)

Penalises large forecasting mistakes more heavily.

Important for operational reliability.

R^2 Score

Measures how well the model explains output variability.

Higher values indicate stronger prediction capability.

MAPE (Mean Absolute Percentage Error)

Measures percentage forecasting error relative to plant capacity.

Commonly used in energy forecasting systems.

Classification Metrics

Precision

Measures how many degradation alerts are actually correct.

Important to minimise false maintenance alarms.

Recall

Measures how many true degradation cases are successfully detected.

Critical for preventing unnoticed efficiency loss.

F1 Score

Balances precision and recall.

Useful for imbalanced degradation datasets.

12. Expected Business Impact

The proposed system is expected to:

- reduce undetected panel degradation
- improve solar generation efficiency
- reduce revenue leakage
- optimise maintenance scheduling
- minimise unnecessary inspections
- improve asset lifespan monitoring
- support proactive maintenance operations

13. Monitoring & Maintenance

Production Monitoring

- daily degradation trend tracking
- rolling performance ratio monitoring
- prediction drift detection
- feature drift analysis
- weather data validation
- latency monitoring

Retraining Triggers

1. Significant drift in residual distributions
2. Degradation detection accuracy decline
3. Seasonal performance degradation
4. Addition of new solar plant types
5. Quarterly scheduled retraining

14. Deployment Architecture

Infrastructure

Component	Technology
Cloud Platform	Azure ML
Batch Processing	Databricks
Storage	Azure Blob Storage
Feature Store	Feast
Monitoring	Grafana + Evidently AI
Streaming	Azure Event Hubs

Serving Strategy

Batch Inference

- daily degradation analysis
- long-term trend calculation
- maintenance reporting

Real-Time Monitoring

- live performance ratio monitoring
- immediate underperformance alerts
- operational dashboard integration

15. Limitations

Known Limitations

1. Extreme weather conditions may affect residual accuracy
2. Heavy cloud variability can create temporary false positives

3. Synthetic degradation data may not fully represent real-world behaviour
4. Sudden catastrophic failures are outside model scope
5. Requires reliable weather and SCADA telemetry

16. Future Improvements

Future enhancements may include:

- Deep learning time-series models
- Satellite-image-based panel inspection
- Drone-assisted thermal imaging integration
- Computer vision crack detection
- Reinforcement learning maintenance optimisation
- Digital twin integration
- Edge AI deployment for solar farms

17. Summary

The SolarShield-Degradation System extends the Omni-Grid renewable energy intelligence platform by introducing advanced panel health monitoring and degradation analytics.

By combining weather-normalised forecasting, residual analysis, rolling trend detection, and machine learning-based degradation scoring, the model enables proactive solar asset management and improves long-term operational efficiency.

The system integrates naturally into the existing Omni-Grid data pipeline and complements the platform's existing:

- Solar Forecasting System
- Inverter Anomaly Detection System
- Predictive Maintenance / RUL System

creating a unified AI-driven renewable energy analytics ecosystem.